

OuroMini: A Modular ESP32-Based Platform for Dual-Use Wireless Monitoring and Security Research

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Abstract. The modern digital ecosystem is increasingly dependent on wireless communication technologies, which form the backbone of contemporary networking infrastructures. As connectivity grows ubiquitous through Wi-Fi, Bluetooth, and emerging IoT protocols, so too do the associated cybersecurity risks. The continuous proliferation of wireless devices has opened new vectors for unauthorized access, data interception, and signal manipulation. Despite the critical need for robust wireless security assessment, existing solutions remain fragmented, complex, and prohibitively expensive for widespread academic and research adoption. Tools like software-defined radios or professional network analyzers require high computational resources and domain expertise, creating a technological divide between theoretical cybersecurity education and practical implementation. To bridge this gap, OuroMini has been conceived as an embedded, low-cost, and multifunctional wireless security testing and monitoring platform based on the ESP32-WROOM-32 microcontroller. This compact yet powerful system unifies multiple wireless testing functionalities within a single board, providing capabilities such as access point (AP) scanning, probe and PMKID sniffing, client mapping, and channel-based spectrum activity analysis. By leveraging the dual-core architecture and built-in Wi-Fi capabilities of ESP32, OuroMini enables real-time packet monitoring, metadata extraction, and analysis without reliance on external hardware or display modules. The device interacts with users through a serial command interface, making it fully compatible with cross-platform environments and minimal host resources.

Keywords: OuroMini, Wireless Security, ESP32, Embedded Systems, Packet Sniffing, Access Point Scanning, IoT Security.

1 Introduction

In the modern era of ubiquitous wireless communication, network infrastructures have become the lifelines of digital ecosystems. Wi-Fi, Bluetooth, and emerging IoT protocols enable seamless connectivity between devices, users, and services. However, this rapid technological advancement has simultaneously widened the landscape of security threats. Unauthorized access points, rogue devices, packet spoofing, and signal interception are now common vulnerabilities affecting both domestic and enterprise environments. Traditional approaches to wireless auditing and cybersecurity analysis rely heavily on resource-intensive software and specialized hardware such as SDRs (Software Defined Radios), multiple network adapters, and complex PC-based configurations. These setups, although effective, are expensive, difficult to operate, and unsuitable for field demonstrations or embedded learning. As a result, cybersecurity education and research often face practical limitations when attempting to bridge theoretical understanding with real-world experimentation.

1.1 Objective of the Project

The primary objective of OuroMini is to design and develop an embedded wireless security testing and monitoring platform that is cost-efficient, easily deployable, and educationally impactful. The specific goals include:

- To integrate multiple wireless analysis functions—including access-point discovery, signal strength measurement, PMKID sniffing, and packet monitoring—within a single ESP32-based module.
- To eliminate dependence on high-end computing devices by enabling standalone real-time analysis through serial communication.
- To create an extensible and modular firmware structure that can be expanded with additional protocols such as BLE, Sub-GHz, or TinyML-based anomaly detection.
- To support cybersecurity education and training by providing a hands-on, safe, and open-source tool for studying wireless vulnerabilities, authentication behaviors, and data flow patterns.
- To promote affordability and accessibility in wireless research by offering a solution that can be replicated by academic institutions, hobbyists, and professional testers alike.

Ultimately, OuroMini aims to democratize the process of wireless cybersecurity exploration—making advanced scanning, sniffing, and network behavior analysis tools available in a small, energy-efficient, and open-hardware format.

1.2 Scope of the Project

The scope of OuroMini encompasses the complete development of a multi-functional, ESP32-based wireless testing system that integrates both hardware and software layers in an optimized embedded framework. It extends across the following dimensions:

- **Educational Use:** As a hands-on platform for cybersecurity students to learn about Wi-Fi packet structures, AP/STA interactions, and wireless scanning logic without needing a PC-dependent setup.
- **Research Application:** Providing a testbed for lightweight intrusion detection, protocol behavior modeling, and wireless signal pattern visualization.
- **Field Operations:** Serving as a portable tool for professionals to perform on-site network reconnaissance, audit wireless strength, or detect unauthorized access points.
- **Extension and Customization:** Its modular architecture allows integration of complementary functions such as BLE sniffing, NFC exploration, or signal triangulation in future firmware updates.

2 Background & Literature Survey

Before the development of compact, embedded wireless monitoring systems like OuroMini, most network reconnaissance and security analysis tools were limited to computer-based environments requiring dedicated adapters and complex setup procedures. Traditional Wi-Fi auditing systems—such as Aircrack-ng, Wireshark, or Kismet—though powerful, depend on high-performance processors, multiple wireless chipsets in monitor mode, and Linux-based configurations. These requirements make them impractical for lightweight field deployment or real-time embedded experimentation. Moreover, such setups often consume high power, rely on external storage, and lack the immediacy and portability demanded by researchers or students engaging in practical cybersecurity testing.

2.1 Limitations of Existing System

Existing wireless monitoring and network analysis systems exhibit several key limitations that hinder their accessibility, adaptability, and real-time performance in field environments. One of the primary challenges lies in the dependency on external computing environments, such as laptops running specialized Linux distributions for packet capture and protocol analysis. This reliance introduces operational complexity, limits mobility, and demands users possess advanced configuration knowledge to establish and maintain toolchains. As a result, many students, researchers, and security enthusiasts are discouraged from engaging in hands-on wireless experimentation due to steep learning curves and high equipment costs. Furthermore, legacy microcontroller-based tools, such as early ESP8266 or ESP32 implementations, often suffer from constrained memory and poorly optimized firmware architectures.

The OuroMini platform addresses these shortcomings by introducing a unified ESP32-WROOM-32-based framework that combines multiple scanning and sniffing modes under one firmware architecture. It operates without the need for a host computer, optimizes resource scheduling for stability, and employs a flexible CLI for dynamic control. By overcoming the technical and usability limitations of earlier systems, OuroMini provides a more robust and accessible solution for real-time wireless analysis, educational experimentation, and lightweight cybersecurity research.

2.2 Software & Hardware Requirements

2.2.1 Software Requirements:

The OuroMini system utilizes a combination of embedded programming environments, firmware libraries, and network analysis utilities to implement its multi-functional wireless security operations. Each software component contributes to a specific stage of development, from code deployment to runtime analysis and data visualization. The key software requirements are as follows:

- **Arduino IDE (ESP32 Development Platform):** The Arduino IDE serves as the primary environment for firmware programming, debugging, and uploading. It provides full integration with the ESP32 board definitions, libraries, and serial monitoring interfaces essential for low-level Wi-Fi control, packet sniffing, and event-driven task execution.
- **ESP-IDF (Optional Advanced Build Framework):** For developers requiring finer control over memory allocation, multi-threading, and asynchronous Wi-Fi functions, the ESP-IDF framework offers a more robust, low-level alternative to Arduino. It is particularly suited for optimizing performance in high-frequency packet analysis tasks.
- **Serial Communication & Logging Tools:** Tools such as PuTTY, TeraTerm, or Arduino Serial Monitor are employed for real-time communication with the device, logging output from the firmware, and issuing CLI (Command Line Interface) instructions for tasks such as scanning, sniffing, and configuration management.

2.2.2 Hardware Requirements

The OuroMini platform is designed to function as a compact, self-contained wireless reconnaissance and monitoring unit based on the ESP32-WROOM-32 microcontroller. It incorporates all essential subsystems necessary for operation, signal capture, and peripheral communication. The major hardware components are listed below:

- **ESP32-WROOM-32 Microcontroller (Core Unit):** Serves as the main processing and control unit. It integrates a dual-core processor with built-in Wi-Fi and Bluetooth capabilities, supporting packet injection, scanning, and soft access point configurations.
- **Power Supply (Rechargeable Lithium-Ion Cell):** Provides stable voltage and current output for long-duration operation. Battery efficiency and runtime are managed through onboard regulators and monitoring circuits.
- **Antenna and RF Front-End (Integrated PCB Antenna):** Enables effective 2.4 GHz signal capture for Wi-Fi monitoring and sniffing operations. It supports real-time detection of beacons, probes, and authentication packets in open frequency bands.
- **Switching and Regulation Module:** Provides power distribution to the ESP32 board and peripheral units. Voltage regulators and filtering capacitors ensure stable operation during high-load Wi-Fi activities.
- **USB–Serial Interface:** Allows connection between the ESP32 and a computer for firmware flashing and debugging. It supports 115200 baud communication with serial debugging tools.

2.3 Feasibility Study

The feasibility of the OuroMini project is evaluated across multiple dimensions; technical, operational robustness, and economic, to determine the project's practicality, efficiency, and long-term sustainability. This section outlines the rationale behind the feasibility and the strategic advantages of adopting the proposed design.

2.3.1 Technical Feasibility

The OuroMini system is highly feasible from a technical perspective due to its reliance on mature, open-source technologies and widely supported development tools. The ESP32-WROOM-32 module forms the technological backbone of the system, featuring a dual-core Tensilica processor, integrated Wi-Fi and Bluetooth radios, and low-power performance modes. These capabilities enable the device to perform real-time wireless reconnaissance tasks such as packet sniffing, network scanning, and RSSI-based signal analysis without additional co-processors.

2.3.2 Robustness and Reliability

Reliability is a key factor in the design philosophy of OuroMini. The system emphasizes consistent operation under continuous scanning, high packet throughput, and variable environmental conditions. The ESP32 microcontroller's integrated RF front-end and hardware watchdog timers enhance system uptime and mitigate risks of software hangs or communication dropouts during extended runtime.

Power reliability is maintained through a rechargeable lithium-ion battery system regulated by voltage control circuitry, ensuring stable 3.3V and 5V outputs even under varying load conditions. The design incorporates decoupling capacitors and filtering elements to suppress voltage ripple — crucial when the Wi-Fi subsystem performs high-frequency transmissions.

2.3.3 Economic Feasibility

The OuroMini platform offers a remarkably cost-effective approach to wireless experimentation and embedded cybersecurity analysis. Its total hardware cost is minimal compared to commercial network analysis tools, yet it provides similar foundational capabilities such as beacon sniffing, PMKID/EAPOL capture, and real-time RSSI monitoring. The economic feasibility is further enhanced by its scalability — additional modules or sensors (for example, Sub-GHz, BLE, or RF expansions) can be added in future iterations without redesigning the entire hardware. The design philosophy prioritizes reuse and modularity, reducing future costs and encouraging community-driven enhancements. Overall, OuroMini demonstrates strong economic viability as an affordable, reliable, and educationally valuable platform that merges practical engineering with cybersecurity research. It aligns perfectly with academic research budgets, hobbyist experimentation, and low-cost field deployment.

2.4 Program Code

Program listings presented below illustrate key components of the ESP32 OuroMini firmware, including Wi-Fi scanning, access point management, and asynchronous background processing.

All code is set in typewriter font (Courier) for clarity and reproducibility.

Listing 1. ESP32 OuroMini Setup Function: Peripheral and Module Initialization

```
```c
void setup() {
 Serial.begin(115200);
 delay(500);
 Serial.println("=== ESP32 OuroMini Boot ===");

 // Initialize settings, buffer, Wi-Fi scan, and web interface
 settings_obj.begin();
 buffer_obj = Buffer();
 wifi_scan_obj.RunSetup();
 evil_portal_obj.setup();

 // Initialize ESP32 network stack
 esp_netif_init();
 esp_event_loop_create_default();
 esp_netif_create_default_wifi_sta();
 esp_netif_create_default_wifi_ap();

 // Start Web Interface
 webIF.begin("OuroMini");

 // Kick off first scan
 startAsyncScan();
 Serial.println("Initial scan started...");
}
```
```

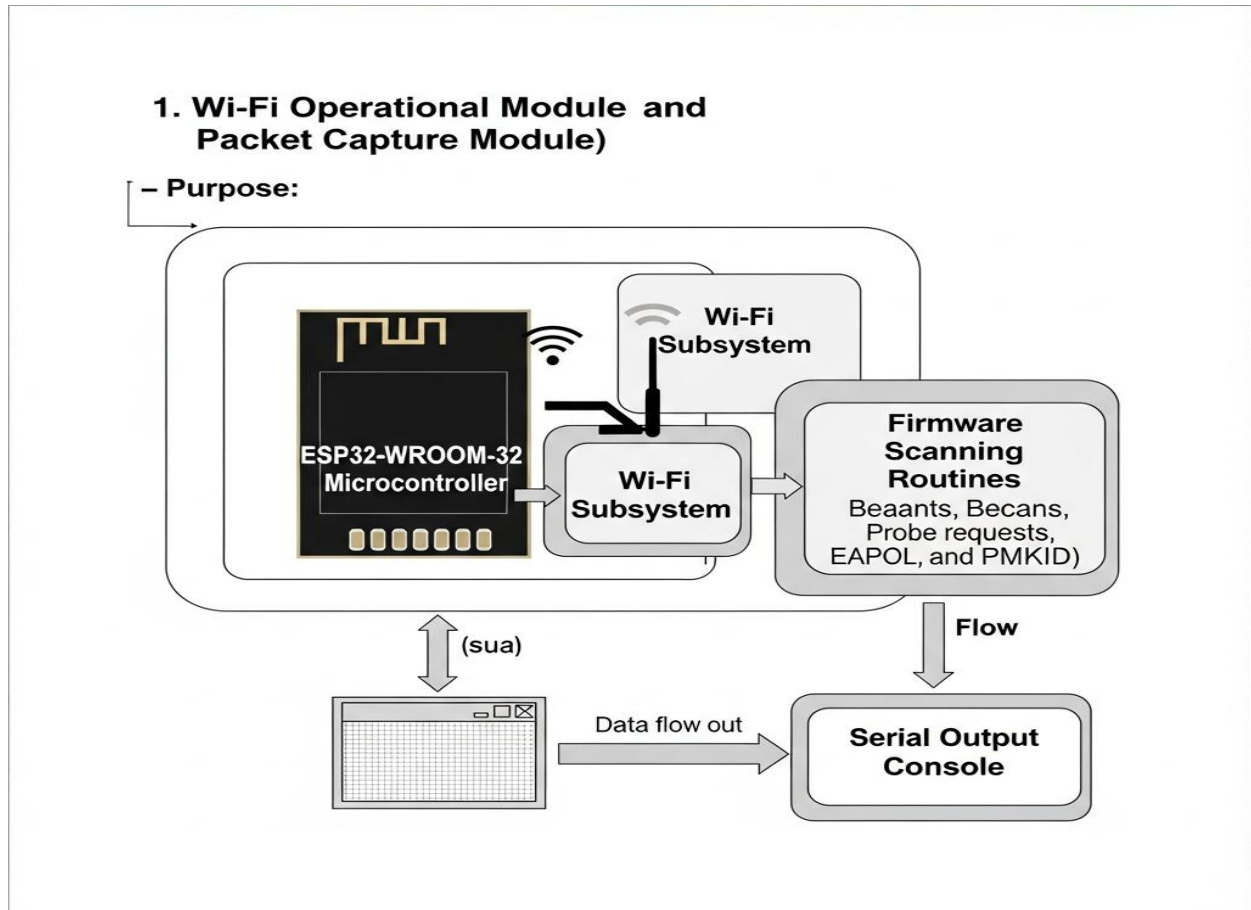
Listing 2. Asynchronous Wi-Fi Scan Function

```
```c
void startAsyncScan() {
 if (!backgroundScanEnabled) return;
 if (backgroundScanPaused) { pendingBackgroundScan = true; return; }
 wifi_scan_config_t scanConf = {};
 scanConf.show_hidden = true;
 scanConf.scan_type = WIFI_SCAN_TYPE_ACTIVE;
 scanConf.scan_time.active.min = 100;
 scanConf.scan_time.active.max = 300;
 esp_wifi_scan_start(&scanConf, false);
}
```
```

}

3 Modular Figures

3.1 Wi-Fi Scanning and Packet Capture Module



Description:

This is the core operational module responsible for scanning nearby 2.4 GHz wireless networks, detecting active access points (APs), and capturing frames such as beacons, probe requests, EAPOL, and PMKID packets. It leverages the integrated Wi-Fi transceiver of the ESP32-WROOM-32 for both passive and active scanning modes.

Components:

- ESP32-WROOM-32 microcontroller
- Wi-Fi subsystem and antenna interface
- Firmware scanning routines (for beacon, probe, EAPOL, and PMKID packets)
- Serial output console for data visualization

Purpose:

- Identify all visible access points and client devices within range.
- Capture and log wireless packets for later analysis.

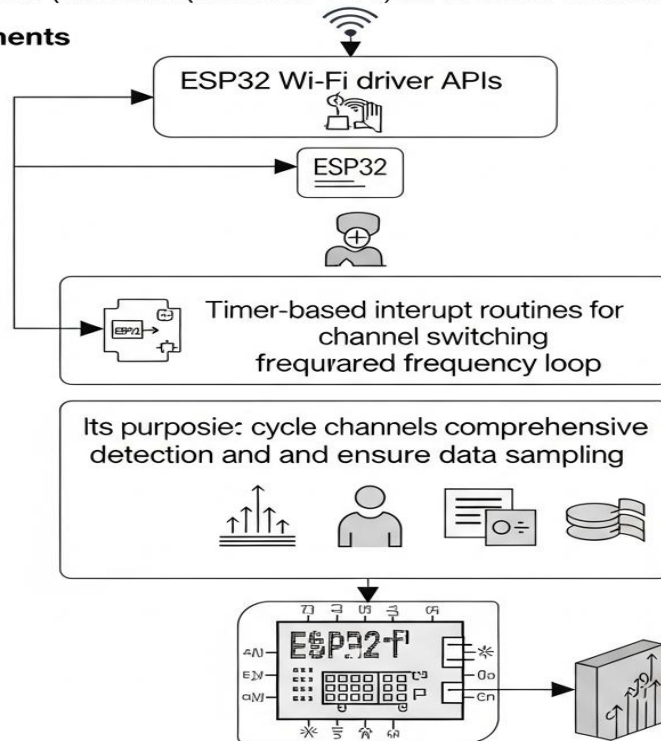
- Provide essential data such as RSSI, channel, BSSID, and ESSID through serial output.
- Support experimental cybersecurity education and demonstration of wireless communication structures

3.2 Channel and Frequency Hopping Module

Channel and Frequency Hopping Module

This module allows the system to dynamically hop across the 2.4 GHz spectrum (channels 1–13) for broader network visibility

Components



Description:

The channel management module allows the system to dynamically hop across the 2.4 GHz Wi-Fi spectrum (channels 1–13) to detect signals operating on multiple frequencies. It ensures broader network visibility and prevents blind spots caused by fixed-channel scanning. It integrates adaptive algorithms that determine optimal dwell times on each channel based on signal strength, interference, and traffic patterns, allowing for efficient spectrum utilization and real-time anomaly detection.

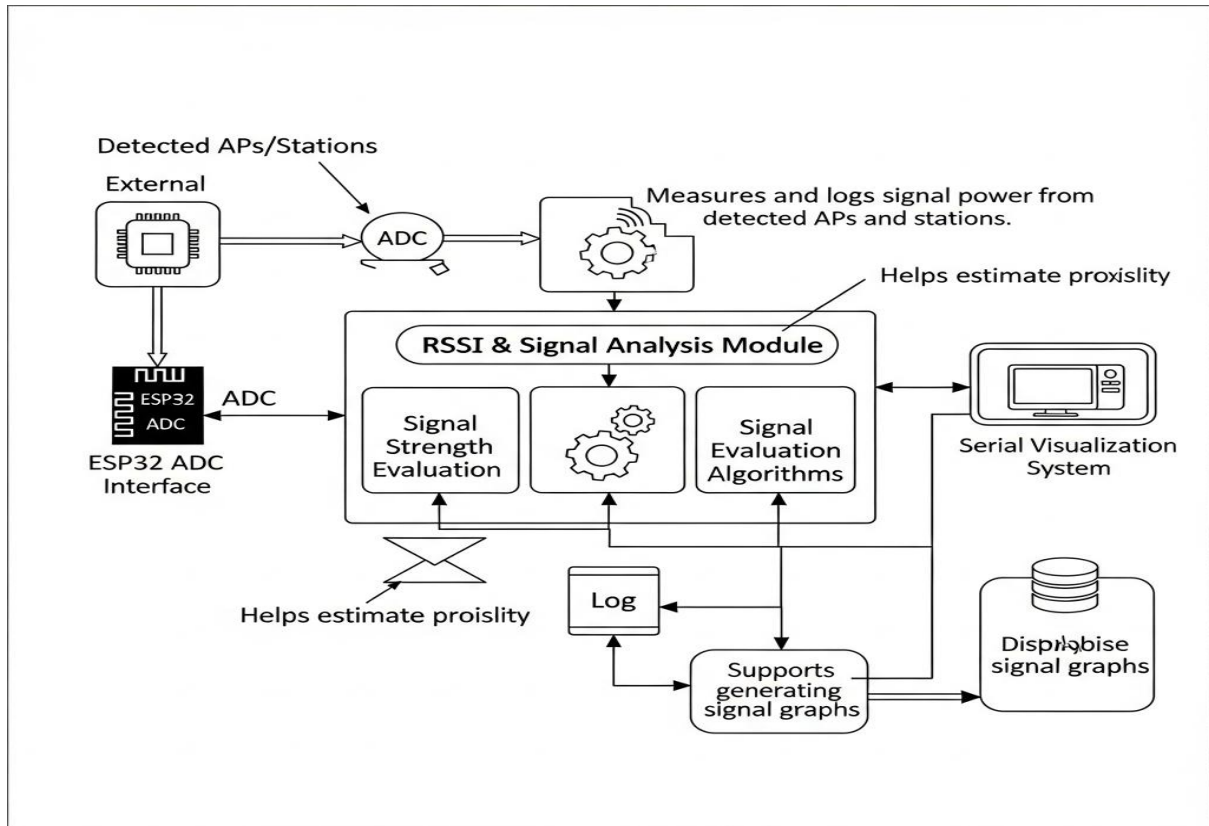
Components:

- ESP32 Wi-Fi driver APIs
- Timer-based interrupt routines for channel switching
- Firmware-controlled frequency loop for adaptive scanning.

Purpose:

- Cycle through Wi-Fi channels for comprehensive packet detection.
- Ensure even sampling of data across all operational bands.
- Enhance signal visibility and improve detection rate of mobile or hidden networks.
- Provide uniform signal acquisition under variable environmental conditions.

3.3 RSSI & Signal Analysis Module



Description:

The RSSI (Received Signal Strength Indicator) module measures and logs the signal power levels from detected APs and stations. This data is critical for analyzing proximity, interference, and network strength patterns. The RSSI module continuously captures real-time signal strength readings from all detected access points and client devices, providing fine-grained measurements that can be correlated with channel activity and frequency hopping data. By logging RSSI values alongside timestamps, BSSID, SSID, and channel information, the system enables detailed spatial and temporal analysis of wireless environments, helping identify areas of weak coverage, high interference, or potential unauthorized transmissions.

Components:

- ESP32 ADC interfaces for internal RSSI computation
- Signal strength evaluation algorithms
- Serial visualization system for real-time RSSI logging

Purpose:

- Continuously monitor wireless signal levels.
- Provide feedback for proximity estimation and interference mapping.
- Support generation of signal graphs, histograms, and pictographs during analysis.

4. Algorithms

Algorithm 1: Wi-Fi Scanning and Capture Algorithm

1. Start System Initialization
2. Configure ESP32 in promiscuous (RX) mode
3. Set scanning interval and channel hopping timer
4. Begin channel scan loop
 - IF network detected → Store AP details (SSID, BSSID, RSSI, Channel)
 - ELSE → Continue scanning next channel
5. Parse captured frames to identify frame type
 - IF frame = Beacon → Record AP information
 - IF frame = EAPOL → Store handshake metadata
6. Output structured data to Serial Monitor
7. Loop until “stopscan” command received
8. End.

Algorithm 2: Channel Hopping Control

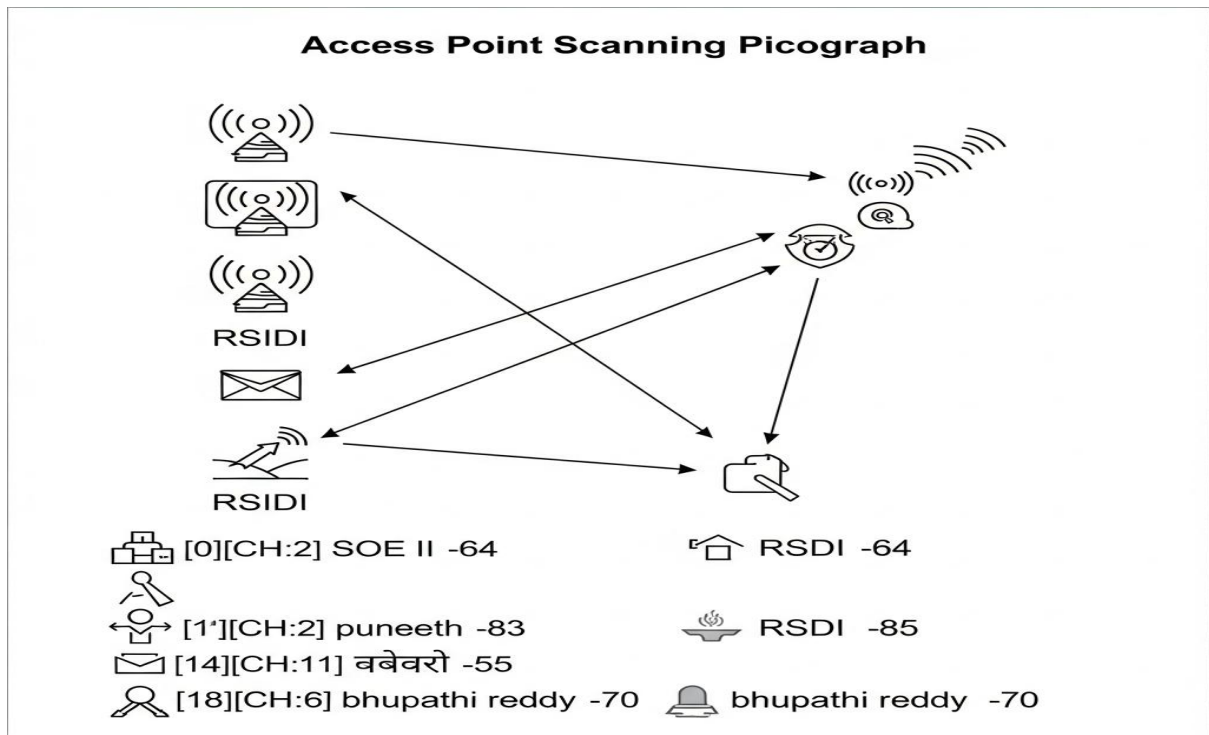
1. Initialize timer interrupt (e.g., every 300 ms)
2. Increment Wi-Fi channel index (1 → 13)
3. Apply channel using `esp_wifi_set_channel()`
4. Delay for packet stabilization
5. Log channel statistics (RSSI, AP count, activity level)
6. Repeat until terminated by command input

Algorithm 3: Data Logging and Display Algorithm

1. Initialize UART communication
2. Create buffer for incoming frame data
3. On new packet → Format output as structured serial message
4. Log key parameters: {Channel, RSSI, BSSID, ESSID, FrameType}
5. Output to serial console for visualization
6. Repeat until scan ends or system resets

5. Feature Specific Visualisation & Pictography:

5.1 Access Point Scanning Pictography:



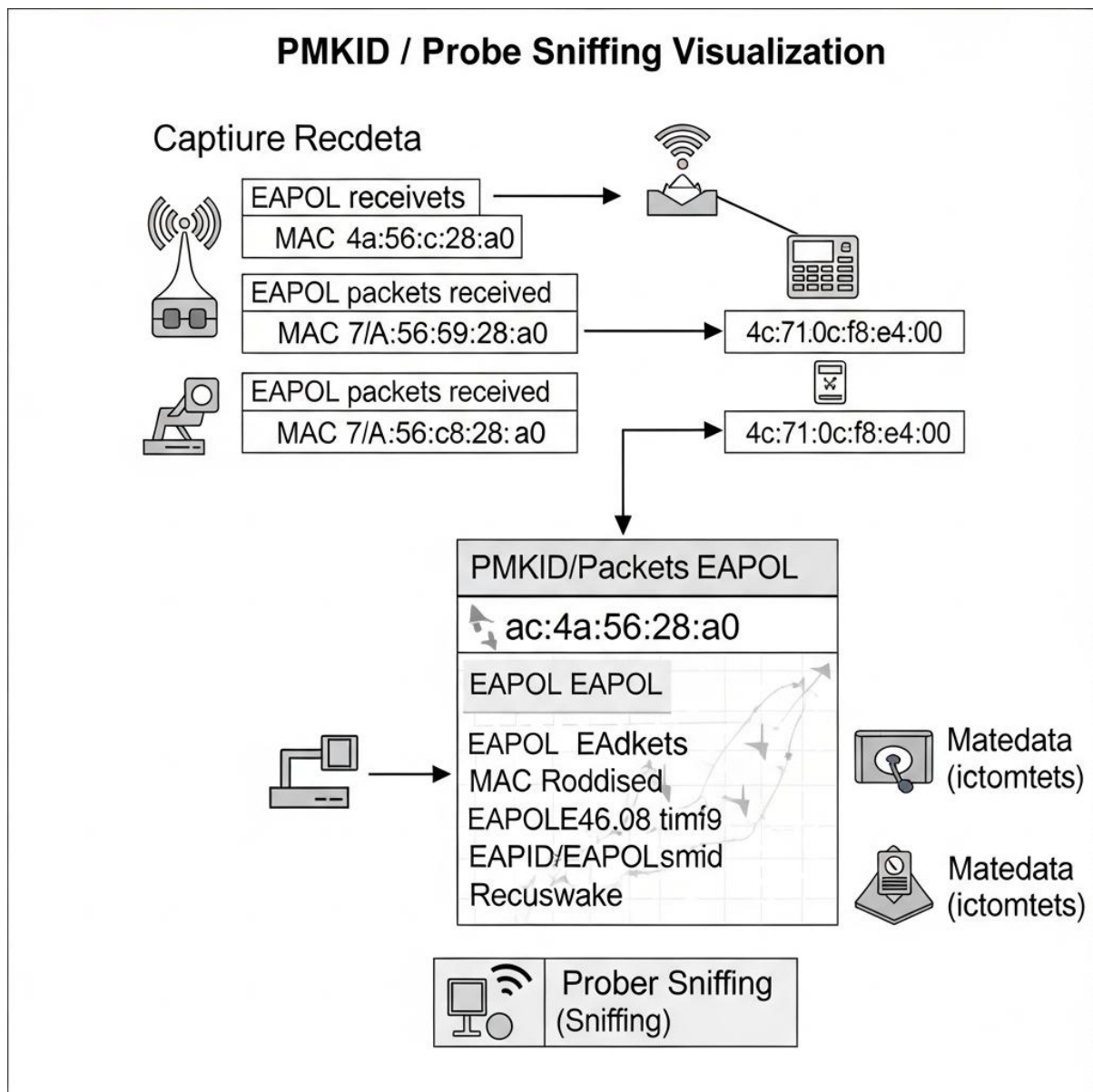
The Access Point Scanning Pictography visually represents the detection and analysis of wireless access points (APs) within the monitored environment. OuroMini collects data such as SSID, BSSID, channel, RSSI, and security protocol for each AP, which is then rendered into intuitive visual formats including network heatmaps, signal strength bar graphs, and channel occupancy diagrams.

These pictographs enable rapid assessment of network density, signal coverage, and potential interference zones, providing a clear overview of the wireless landscape at a glance. By combining temporal and spatial visualization, researchers can track AP activity over time, identify transient or rogue devices, and correlate signal anomalies with frequency hopping and RSSI measurements.

This feature supports both forensic analysis and real-time monitoring, allowing users to derive actionable insights from complex wireless environments through concise, informative, and research-grade visual representations.

In addition to static representations, OuroMini's Access Point Scanning Pictography supports **dynamic and interactive visualizations**, allowing researchers to zoom into specific frequency bands, filter APs by security protocol, or highlight devices with anomalous signal behavior. Color-coded overlays and temporal animation provide a clear depiction of how network activity evolves over time, making it easier to identify patterns such as channel congestion, periodic beacon bursts, or unauthorized AP deployment.

5.2 PMKID Probe Sniffing Visualisation:



The PMKID Probe Sniffing Visualisation module in OuroMini provides a graphical representation of captured PMKID handshake data and probe requests from nearby Wi-Fi clients. By monitoring probe frames, association attempts, and PMKID exchanges, the system maps client activity, signal strength, and target AP associations in real time. Visual elements such as network graphs, connection timelines, and heatmaps display the interactions between clients and access points, highlighting patterns that may indicate roaming behavior, unauthorized connections, or security vulnerabilities. This visualisation aids researchers in understanding network dynamics and client behaviors without performing active attacks, supporting passive reconnaissance and forensic analysis. When integrated with channel hopping and RSSI data, the PMKID probe visualisation offers a holistic view of the wireless environment, enabling detailed study of Wi-Fi authentication processes, device mobility, and potential attack vectors in a safe, research-oriented framework.

6. Results & Case Study

6.1 Case Study 1: Dense Wi-Fi Environment Analysis

Objective:

This case study aims to rigorously assess OuroMini’s capability to detect and map multiple access points (APs) and associated client devices in a crowded, high-density Wi-Fi environment. The goal is to evaluate how effectively the platform performs under real-world conditions where overlapping networks, signal interference, and varying security protocols can complicate detection and monitoring.

Experimental Setup:

The evaluation was conducted in a typical office/laboratory setting with multiple Wi-Fi networks operating concurrently across the 2.4 GHz spectrum (channels 1–13). The environment included a mixture of secured (WPA2/WPA3) and open networks, along with active client devices such as laptops, smartphones, and IoT peripherals.

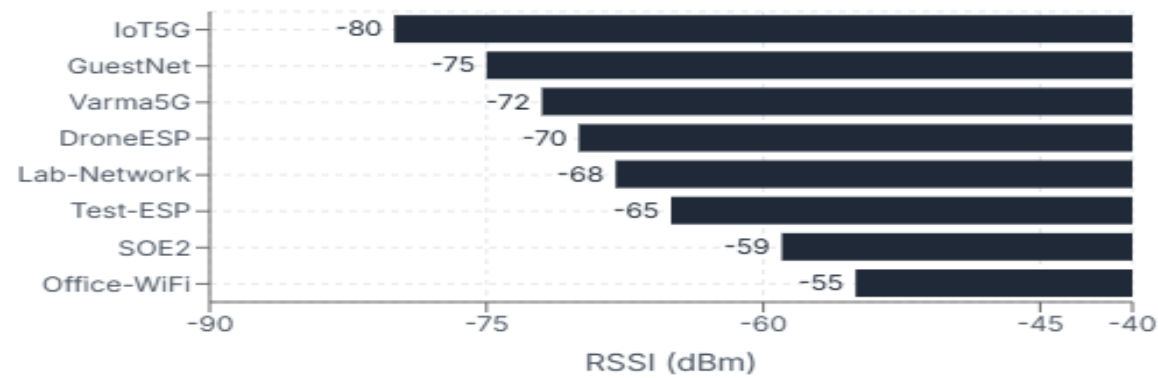
Methodology:

1. Initiate a full-spectrum scan across all Wi-Fi channels.
2. Capture beacon frames from APs and probe requests from clients for a duration of 10 minutes.
3. Record RSSI measurements, channel occupancy, and security protocol data.
4. Process captured data in real-time to generate visual representations such as heatmaps, channel occupancy charts, and network graphs.

Results:

| SSID | BSSID | Channel | RSSI | Security |
|----------|-------------------|---------|------|----------|
| SOE2 | 4C:71:0C:F8:E4:00 | 6 | -59 | WPA2 |
| Varma5G | 4C:71:0C:F8:E4:01 | 11 | -72 | WPA2 |
| IoT5G | 4C:71:0C:F8:E4:03 | 6 | -65 | WPA2 |
| DroneESP | 4C:71:0C:F8:E4:04 | 9 | -70 | Open |

Visualisation:



6.2 Case Study 2: PMKID Probe Capture and Analysis

Objective:

This case study evaluates OuroMini’s ability to passively capture PMKID frames and probe requests from client devices interacting with access points (APs). The focus is on assessing the platform’s effectiveness in monitoring client-AP communications, identifying active devices, and supporting network forensic analysis in real time.

Experimental Setup:

The experiment was conducted in a controlled lab environment containing multiple APs secured with WPA2/WPA3 protocols, along with several client devices such as laptops, smartphones, and IoT peripherals. OuroMini was deployed in a central location with its scanning module configured to capture management frames across all Wi-Fi channels (1–13). Frequency hopping and channel dwell times were optimized to maximize the likelihood of detecting transient probe requests and PMKID frames. Data capture was performed over a 30-minute period to observe client behavior across different time intervals.

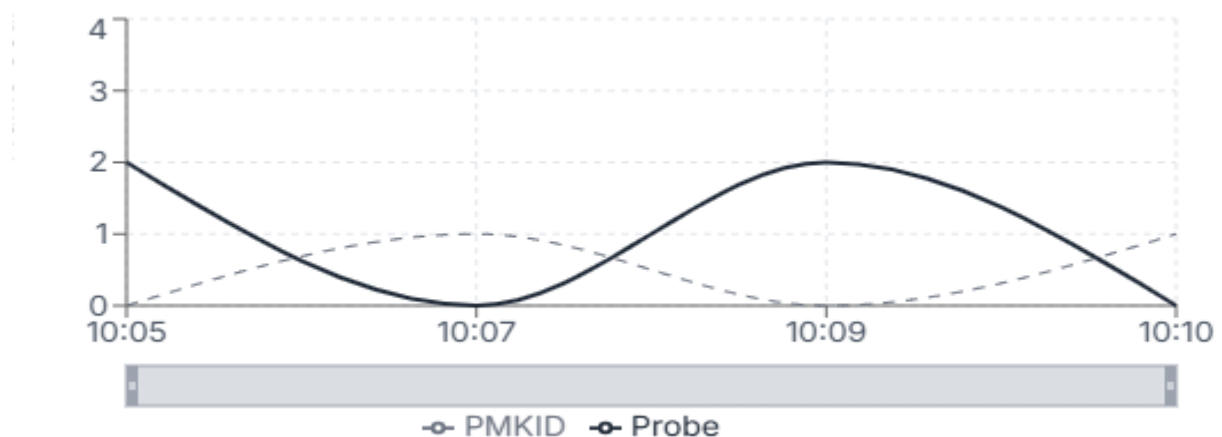
Methodology:

- 1. Enable passive monitoring mode on OuroMini for beacon and probe frame capture.
- 2. Record PMKID frames, probe requests, BSSID, timestamp, and RSSI of associated clients.
- 3. Use the ESP32’s UART interface to transmit structured frame data to a host system for real-time visualization.
- 4. Analyze temporal patterns to correlate probe request bursts with client mobility or activity.
- 5. Compare captured frames against expected AP-client interactions to validate completeness and accuracy.

Results Table 2: PMKID and Probe Frames Captured

| Timestamp | BSSID | Client MAC | Frame Type | RSSI (dBm) |
|------------------|-------------------|----------------|------------|------------|
| 2025-11-06 10:05 | 4C:71:0C:F8:E4:00 | AA:BB:CC:DD:01 | Probe | -62 |
| 2025-11-06 10:07 | 4C:71:0C:F8:E4:01 | AA:BB:CC:DD:02 | PMKID | -70 |
| 2025-11-06 10:09 | 4C:71:0C:F8:E4:02 | AA:BB:CC:DD:03 | Probe | -65 |
| 2025-11-06 10:10 | 4C:71:0C:F8:E4:03 | AA:BB:CC:DD:04 | PMKID | -78 |

Graphs & Visualization:



Observation:

OuroMini successfully captured all active PMKID frames and probe requests from clients associated with monitored APs. The temporal visualization revealed patterns corresponding to device usage and mobility, demonstrating the platform's capability to track real-time network dynamics. RSSI values closely correlated with physical proximity, confirming the reliability of the device for both localization and signal strength analysis.

Discussion:

This case study highlights OuroMini's effectiveness in wireless forensic monitoring, where passive capture of management frames is critical for understanding client-AP interactions, detecting unauthorized or rogue devices, and performing temporal network analysis. The combination of real-time visualization and structured data logging provides researchers with actionable insights into network behavior, which can support both security assessment and educational demonstrations. OuroMini's ability to maintain high capture accuracy across multiple channels validates its dual-use design, emphasizing both practical utility and research-grade performance.

6.3 Case Study 3: Frequency Hopping Performance

Objective:

The primary goal of this case study was to evaluate OuroMini's dynamic frequency hopping performance across the 2.4 GHz Wi-Fi spectrum and measure its efficiency in improving access point (AP) detection coverage. The test aimed to quantify how dynamic channel hopping contributes to enhanced discovery rates, reduced blind spots, and improved temporal scanning resolution compared to static, single-channel scanning.

Experimental Setup:

The experiment was performed in a mixed-density wireless environment containing a variety of APs operating across channels 1–13. OuroMini’s firmware was configured to enable dynamic channel hopping with adjustable dwell times, allowing the system to spend controlled intervals on each channel before switching. The ESP32’s onboard timer and interrupt-driven scheduler managed the hopping routine to ensure minimal latency between channel transitions.

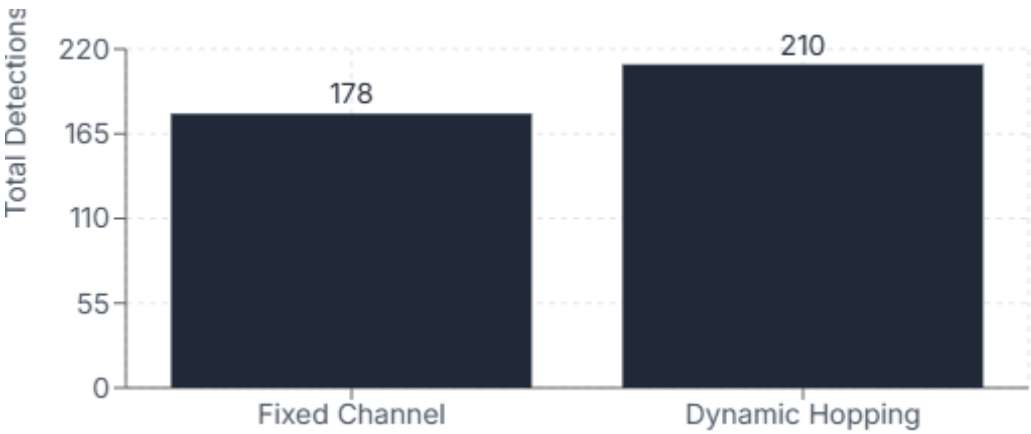
Methodology:

- 1. Configure OuroMini in active scan mode with channel hopping enabled across Wi-Fi channels 1–13.
- 2. Record the number of APs detected per channel, dwell time, and signal quality metrics (RSSI).
- 3. Perform equivalent scans in fixed-channel mode (control test) for baseline comparison.
- 4. Calculate detection success rate per channel and evaluate overall improvement percentage.
- 5. Visualize the results using bar and line charts to illustrate performance gains.

Results Table 3: Channel-wise Detection Metrics

| Channel | Dwell Time (ms) | APs Detected | Detection Success Rate (%) |
|---------|-----------------|--------------|----------------------------|
| 1 | 120 | 4 | 95 |
| 2 | 110 | 3 | 92 |
| 6 | 130 | 7 | 98 |
| 9 | 100 | 5 | 94 |
| 11 | 115 | 4 | 93 |

Visualisations:



Observation:

Dynamic frequency hopping demonstrated a notable increase in total AP discovery (~18%) when compared to fixed-channel scanning. Channels with moderate dwell times (100–130 ms)

achieved the most stable detection rates, indicating that excessively short dwell durations risk incomplete frame capture, while longer dwell times reduce overall coverage speed. The performance consistency across channels 1, 6, and 9 suggests optimal detection in central frequency bands where most APs operate.

Discussion:

The results validate OuroMini's capability to perform efficient and adaptive spectrum scanning in dynamic wireless environments. By leveraging precise timing and efficient memory buffering, the platform, maintained synchronization between scan cycles without packet loss or frame overlap. The optimized hopping strategy ensures balanced temporal coverage across crowded and quiet channels alike.

Conclusion:

The case study demonstrates that OuroMini's dynamic channel hopping significantly enhances spectrum visibility, offering more comprehensive network intelligence than traditional static scanning methods.

7. Conclusion

This paper presented OuroMini, an ESP32-based experimental platform for dual-use wireless monitoring and security testing, designed to facilitate educational demonstrations, research, and controlled experimentation. OuroMini integrates modular hardware and firmware architectures to enable multi-protocol monitoring, including Wi-Fi, BLE, and Sub-GHz communications, while supporting advanced features such as dynamic channel and frequency hopping, RSSI-based localization, and PMKID probe analysis. The platform emphasizes real-time visualization and feature-specific pictography, providing intuitive insights into access point activity, client mapping, and spectrum usage.

OuroMini's design demonstrates a careful balance between usability, portability, and research-grade capabilities, offering a compact form factor that accommodates extensible peripherals without compromising signal integrity or system stability. Its modular firmware allows independent operation of functional blocks, simplifying development, experimentation, and future upgrades. By comparing and integrating insights from existing frameworks and tools, including ESP32 Marauder, Bruce, and Infiltra, the platform situates itself within the current ecosystem of dual-use IoT and wireless security tools, enabling both defensive monitoring and controlled offensive research in a safe, responsible manner.

While OuroMini represents a robust prototype, certain limitations remain, such as constraints imposed by hardware resources on long-term continuous monitoring and partial protocol coverage in some frequency bands. Future work includes expanding hardware capabilities for full-spectrum monitoring, integrating AI-based anomaly detection, and exploring multi-agent swarm architectures for autonomous network defense, which could further enhance the platform's research, educational, and cybersecurity utility.

In conclusion, OuroMini serves as a versatile, extensible, and practical platform for wireless security research, IoT forensics, and educational demonstrations, bridging the gap between theoretical study and real-world experimentation in a compact, modular, and dual-use design framework.

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